

Foundations of Artificial Intelligence

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Exercise Sheet 9 — Solutions

Exercise 9.1 (1+1 marks)

We investigate phase transitions for arbitrary CNF formulas with n variables, m clauses, and k variables in each clause.

If we vary m while fixing n and k , then large values for m represent an *over-constrained region* since all else being equal having more clauses means having more subformulas connected by a logical and, which constrains the formula more.

For the following variations, do large values for the non-fixed variable lead to an under-constrained or over-constrained region? Justify your answer.

- (a) Varying k while fixing n and m .
- (b) Varying n while fixing m and k .

Solution:

- (a) With a fixed number of variables and clauses, having larger clauses in each clause makes each clause easier to satisfy because there are more options, meaning large values of k represent an *under-constrained* region.
- (b) When the number and size of clauses is fixed, having more available variables means that each variable occurs in less clauses on average, meaning each variable is less constrained, thus resulting in an *under-constrained* region.

Exercise 9.2 (2+1 marks)

Consider the following scenario: it is June 24, 2026, the day of the “Foundations of Artificial Intelligence” exam. You just finished studying for the exam together with your friends at the DMI and have a bit of time to spare before the exam. To relax your mind, you want to go swimming in the Rhine and eat an ice cream. Swimming in the Rhine is possible starting at the *Tinguely museum* bringing you back to *Mittlere Brücke* close to the DMI. In the end, you must arrive at the *Organische Chemie* to write the exam. Ice cream can be bought at the *Mittlere Brücke* or close to the *Organische Chemie*.

- (a) Formalize the above scenario as a STRIPS planning task adhering to the following requirements:
 - There are locations *Mittlere Brücke*, *Organische Chemie* and *Tinguely museum*.
 - Initially, you are located at *Mittlere Brücke* (the DMI).
 - Your goal is to be located at *Organische Chemie*, cooled down from swimming in the Rhine and having eaten an ice cream.
 - You can move freely between any pair of locations.
 - You can swim from *Tinguely museum* to *Mittlere Brücke* but not between any other pair of locations.
 - You can enjoy ice cream at *Mittlere Brücke* or at the *Organische Chemie* but not at *Tinguely museum*.
 - Choose a reasonable cost function for the action costs and briefly explain what the costs represent and why they are reasonable in this scenario.

- (b) Provide an optimal plan for your planning task together with its cost.

Solution:

- (a) Let $L = \{MB, OC, TM\}$ be the set of locations denoting *Mittlere Brücke*, *Organische Chemie* and *Tinguely museum*, respectively. We define $\Pi = \langle V, I, G, A \rangle$ where

- $V = \{at-l \mid l \in L\} \cup \{went-swimming, enjoyed-ice-cream\}$
- $I = \{at-MB\}$
- $G = \{at-OC, went-swimming, enjoyed-ice-cream\}$
- $A = \{go-from-l-to-l' \mid l, l' \in L, l \neq l'\} \cup \{swim\} \cup \{eat-ice-cream-at-l \mid l \in \{MB, OC\}\}$ with the following preconditions, effects and costs:

a	$pre(a)$	$add(a)$	$del(a)$	$cost(a)$
$go-from-l-to-l'$	$\{at-l\}$	$\{at-l'\}$	$\{at-l\}$	$12 \cdot dist(l, l')$
$swim$	$\{at-TM\}$	$\{at-MB, went-swimming\}$	$\{at-TM\}$	30
$eat-ice-cream-at-l$	$\{at-l\}$	$\{enjoyed-ice-cream\}$	$\emptyset$	5

$$\text{where } dist(l, l') = \begin{cases} 1 & \text{if } \{l, l'\} = \{MB, OC\} \\ 2 & \text{if } \{l, l'\} = \{MB, TM\} \\ 3 & \text{if } \{l, l'\} = \{OC, TM\} \end{cases}$$

Since we need to make sure we're at the *Organische Chemie* in time for the exam, we chose our action costs to resemble the amount of time it takes to execute the action in minutes. We estimate how long it takes to get from one location to another based on the walking distance between the two represented by the function $dist$: 1km between *Mittlere Brücke* and *Organische Chemie*, 2km between *Mittlere Brücke* and *Tinguely Museum*, and 3km between *Organische Chemie* and *Tinguely Museum*. Assuming a walking speed of $5 \frac{\text{km}}{\text{h}}$, we get the cost estimates of $12 \cdot dist(l, l')$. Swimming probably takes around 30 minutes and eating ice cream should be doable in 5 minutes. Knowing how much time is left until the exam, the cost of a plan then indicates whether it's feasible to execute it before the exam, or whether we have to give up ice cream and/or Rhine swimming.

- (b) An optimal plan is $\pi = \langle go-from-MB-to-TM, swim, eat-ice-cream-at-MB, go-from-MB-to-OC \rangle$ with $cost(\pi) = 71$.

Exercise 9.3 (0.5+1+1+1.5 marks)

Consider the following variant of the *Seven bridges of Königsberg* problem: for a given set of bridges that cross a river, is there a tour that starts and ends at the same location and crosses each bridge exactly once? Formally, the problem can be defined as follows: given a graph $G = (V, E)$ with a set of vertices V and a set of edges $E \subseteq V \times V$ and an initial vertex $v_0 \in V$, is there a sequence of vertices from V such that i) all pairs of subsequent vertices are connected by an edge from E , ii) each edge in E occurs exactly once in the sequence, and iii) the first and last vertex of the sequence is v_0 ? For an illustration of the problem, consider the following examples:



The initial vertex is v_0 in both cases. For the graph on the left side, there is such a tour, e.g., $\langle v_0, v_1, v_2, v_0 \rangle$, while there is no such tour for the graph on the right side.

On ADAM, you can find more resources required to solve this exercise in the file `koenigsberg.zip`.

- (a) In `koenigsberg.zip`, you find a PDDL description of the (original instance of the) *Seven bridges of Königsberg* problem. The domain description (variables and actions) is given in the file `bridges-once-domain.pddl`, and the problem description (objects, initial state and goal description) is given in the file `koenigsberg-problem.pddl`. Provide a graphical representation of the problem in the same way as the example above. Please do not forget to mark which location is the initial location.

- (b) Obtain the domain-independent planning system *Fast Downward* from

<https://www.fast-downward.org/latest/releases/24.06/>.

Use *Fast Downward* with a configuration that performs greedy best-first search with the delete relaxation heuristic FF to solve the Seven bridges of Königsberg problem from `koenigsberg.zip`. To do so, invoke the planner with

```
./fast-downward.py bridges-once-domain.pddl koenigsberg-problem.pddl
--search "eager-greedy([ff()])"
```

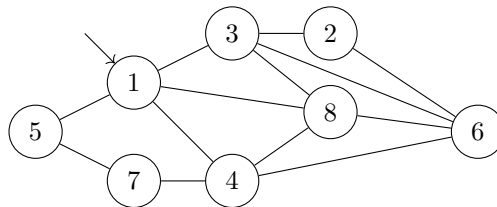
Provide the runtime and the number of expanded states. Discuss whether the solution is optimal. Is the problem solvable? If it is, provide the plan that was found.

- (c) Modify the domain description (`bridges-once-domain.pddl`) such that it is possible to use the same bridge more than once. Solve the resulting problem with the same Fast Downward configuration that was used in part (b) of this exercise.

Provide the runtime and the number of expanded states. Is the problem solvable? If it is, provide the plan that was found.

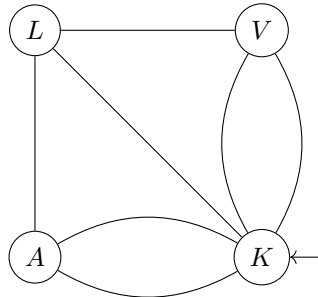
- (d) Formalize the following instance of the bridges domain in PDDL and solve it with the same Fast Downward configuration that was used in part (b) of this exercise. Use the original domain `bridges-once-domain.pddl`, where each bridge may be traversed only once.

Provide the runtime and the number of expanded states. Discuss whether the solution is optimal. Is the problem solvable? If it is, provide the plan that was found.



Solution:

- (a) In the graph below, *A* stands for Altstadt, *V* for Vorstadt, *K* for Kneiphof and *L* for Lomse:



- (b) The problem was detected unsolvable within $\sim 1\text{ms}$, for which 45 states were expanded. Since the planner detects unsolvability, the solution is optimal.

- (c) In order to allow bridges to be traversed several times, it is sufficient to remove the effect `(not (bridge-has-not-been-used ?b))`. With the altered domain the planner finds the following solution with 10 expansions in ~ 1 ms:

```
cross-bridge koettelbruecke kneiphof vorstadt (1)
cross-bridge gruene-bruecke vorstadt kneiphof (1)
cross-bridge schmiedebruecke kneiphof altstadt (1)
cross-bridge holzbruecke altstadt lomse (1)
cross-bridge honigbruecke lomse kneiphof (1)
cross-bridge gruene-bruecke kneiphof vorstadt (1)
cross-bridge hohe-bruecke vorstadt lomse (1)
cross-bridge holzbruecke lomse altstadt (1)
cross-bridge kraemerbruecke altstadt kneiphof (1)
```

- (d) The planner finds the following solution with 14 expansions in ~ 1 ms:

```
cross-bridge b11 l1 l4 (1)
cross-bridge b10 l4 l7 (1)
cross-bridge b8 l7 l5 (1)
cross-bridge b6 l5 l1 (1)
cross-bridge b4 l1 l3 (1)
cross-bridge b12 l3 l6 (1)
cross-bridge b1 l6 l2 (1)
cross-bridge b7 l2 l3 (1)
cross-bridge b3 l3 l8 (1)
cross-bridge b13 l8 l6 (1)
cross-bridge b2 l6 l4 (1)
cross-bridge b9 l4 l8 (1)
cross-bridge b5 l8 l1 (1)
```

Since every bridge must be traversed exactly once, all plans are optimal.

Exercise 9.4 (1 marks)

Specify a STRIPS planning task Π which is unsolvable but whose delete relaxation Π^+ is solvable.

Solution:

Let $\Pi = \langle \{a, b\}, \{a\}, \{a, b\}, \{a_1, a_2\} \rangle$ with $pre(a_1) = \{a\}$, $add(a_1) = \{b\}$, $del(a_1) = \{a\}$, $cost(a_1) = 1$ and $pre(a_2) = \{b\}$, $add(a_2) = \{a\}$, $del(a_2) = \{b\}$, $cost(a_2) = 1$. Π is unsolvable since whenever we achieve a or b we delete the other one, but in Π^+ we have that $\langle a_1^+ \rangle$ is a plan since a_1^+ does not have any delete effects.